AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM341
Computer Vision

Final Examination
Summer Semester 2025/2026

Time Allowed: 3 Hours

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Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART – 1 Total Marks 10

Question 1: Select the correct choice for each of the following statements.

1. Which operation reduces image size while preserving important features?
   A) Convolution
   B) Pooling
   C) Thresholding
   D) Histogram
2. Which filter is best for removing salt-and-pepper noise?
   A) Gaussian
   B) Mean
   C) Median
   D) Laplacian
3. PCA is mainly used for:
   A) Classification
   B) Feature extraction / dimensionality reduction
   C) Edge detection
   D) Segmentation
4. Which transformation preserves parallel lines?
   A) Homography
   B) Perspective transform
   C) Affine transform
   D) Projective warp
5. Which activation function is most commonly used in CNN hidden layers?
   A) Sigmoid
   B) ReLU
   C) Softmax
   D) Tanh

PART – 2 Total Marks 15

Question 2: (5 marks)

a) Name two feature detection algorithms used in computer vision.
b) Give two applications of feature matching.

Question 3: (5 marks)

a) Name two optical flow methods used for motion estimation.
b) Give two applications of optical flow.

Question 4: (5 marks)

a) What is a disparity map?
b) Complete the following Python code to compute the disparity map using OpenCV.

```
import cv2
import numpy as np
left_img = cv2.imread('left.png', cv2.IMREAD_GRAYSCALE)
right_img = cv2.imread('right.png', cv2.IMREAD_GRAYSCALE)
```

a)
Create stereo block matching object
stereo = _____

b) Compute disparity
disparity = _____

c) Normalize disparity for display
disp_norm = _____

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PART – 3 Total Marks 25

Question 5: (6 marks)

Describe the architecture and function of the following CNN model:

```
model = models.Sequential([

layers.Conv2D(32,(3,3),activation='relu',input_shape=(64,64,3)),
    layers.MaxPooling2D((2,2)),
    layers.Flatten(),
    layers.Dense(64,activation='relu'),
    layers.Dense(5,activation='softmax')
])
```

Question 6: (6 marks)

a)Define **K-Nearest Neighbors (KNN)**.
What are its types?
b)Given:
X = data.drop('label', axis=1)
y = data['label']

Write Python code to:

1) Split data into train/test
2) Train KNN with k=3
3) Predict output for input I
4) Predict the output by applying the model in part 2 on new input I

Question 7: (6 marks)

a) What is YOLOv5?
b) Why is YOLOv5 suitable for real-time object detection?

Question 8: (7 marks)

a) List three applications of dimensionality reduction.
b) Write Python code to:

1. Create dataset with 4 samples and 5 features
2. Apply PCA with 3 components
3. Fit model
4. Print transformed data

End of questions

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